

HERITAGE SHIELD

AI-Powered Predictive Conservation & Digital Twin Platform for India's Cultural Heritage

One-line pitch: A living digital intelligence system that creates a digital twin of heritage assets, assesses their condition and surrounding risks, predicts deterioration before irreversible damage occurs, and tells authorities what needs attention, where, when, and why.

1. Executive Summary

India's heritage is exposed not only to natural disasters, but to continuous structural, environmental and human pressures. Conservation is often periodic and reactive: damage becomes visible, an inspection is commissioned, and intervention follows. Heritage Shield shifts the model to predictive and preventive conservation by combining computer vision, GIS, historical records, environmental data, time-series analysis and digital twins.

The platform creates a dynamic digital profile for each heritage asset, tracks how it changes over time, identifies likely causes of deterioration, calculates a multidimensional Heritage Health/Vulnerability Index, forecasts deterioration and recommends prioritized interventions.

2. The Strategic Positioning

Avoid positioning as	Position as
"A disaster-management system for heritage sites."	"A predictive conservation and heritage intelligence system."
Disaster response as the main problem.	Preventing irreversible heritage loss as the main problem.
Static digitisation / 3D documentation.	Living digital twins that change with new evidence.
Risk detection only.	Detect → predict → prioritize → recommend intervention.

3. Core Problem

Heritage information is fragmented across archaeological records, photographs, conservation reports, GIS datasets, archival material, environmental observations and local knowledge. The challenge is not simply a lack of data; it is the lack of a unified system that converts that data into continuous, actionable intelligence.

Key questions the system should answer: What has changed? Which component is deteriorating fastest? Why is it deteriorating? What is likely to happen next? Which asset/component should be inspected first? What intervention should be prioritized, and how urgently?

4. What Heritage Shield Builds

Dynamic Heritage Profile: Historical identity, architectural period, cultural significance, materials, condition and inspection history.

Digital Twin: A structured 2D/3D representation of the site, ideally down to important architectural components.

AI Visual Assessment: Detects cracks, discoloration, vegetation growth, surface damage, moisture indicators and other visible anomalies.

Change Detection: Compares photographs/scans over time to identify new or worsening deterioration.

Heritage Health Index: Combines structural, material, environmental, human-pressure and hazard indicators into an interpretable score.

Predictive Engine: Uses historical observations and risk factors to forecast deterioration trends and flag emerging problems.

Risk & Scenario Engine: Models environmental and hazard exposure, including flood, earthquake, landslide, fire and extreme-weather scenarios.

Intervention Engine: Converts detected risks into prioritized inspection and conservation recommendations.

Heritage Command Dashboard: Provides authorities with site-level and regional prioritization, maps, trends and alerts.

5. Data & Intelligence Pipeline

Stage	Inputs / Processing	Output
OBSERVE	Images, drone/LiDAR where available, GIS, historical records, written/visual/aural data, surveys	Enhanced/annotated data, context
DIGITISE	Structure and component modelling	Digital twin / structured heritage profile
ASSESS	Computer vision + expert validation	Damage/anomaly detections
TRACK	Historical/current comparisons + time series	Deterioration trends
PREDICT	Risk factors + condition history + ML/analytics	Forecasted deterioration / vulnerability
PRIORITISE	Health, urgency, significance, risk reduction and resources	Ranked intervention priorities
ACT	Conservation recommendations + inspection tasks	Action plan
LEARN	Post-intervention observations	Updated model and evidence

6. Heritage Health & Vulnerability Model

A central product concept is an interpretable Heritage Health Index. It should not be a black-box number; the dashboard should expose the factors driving the score. Example dimensions include:

Dimension	Examples
Structural condition	Cracks, deformation, stability indicators
Material condition	Stone/brick/plaster decay, surface loss
Environmental stress	Temperature, humidity, rainfall, pollution
Moisture risk	Water ingress, dampness, drainage-related indicators
Human pressure	Traffic vibration, construction, tourism, vandalism, encroachment
Climate / hazard exposure	Flood, earthquake, landslide, fire, extreme weather
Historical significance	Cultural/architectural importance used for prioritization

7. The Key Innovation: From Detection to Decision

Traditional workflow: Document → Inspect periodically → Discover damage → Respond.

Heritage Shield workflow: Document → Monitor → Detect change → Understand drivers → Predict deterioration → Prioritize → Intervene → Measure outcome.

The system therefore aims to become a decision-support layer for conservation authorities, not another static heritage database.

8. Users

User	Primary value
Government / heritage authorities	Regional monitoring, prioritization, inspection planning, conservation resource allocation
Archaeologists / conservationists	Detailed site condition, evidence comparison, annotations and intervention tracking
Researchers	Structured heritage, environmental and deterioration datasets
Citizens / visitors	Damage reporting, heritage discovery and participatory monitoring

9. SIH MVP — What We Actually Build

The team should resist building a national-scale system during the hackathon. The strongest MVP is a polished, end-to-end demonstration using one or two representative heritage sites and a controlled set of historical/current imagery and contextual data.

- Interactive India/site map and heritage profile.
- Digital twin or 3D/2D site representation.
- Upload and compare historical/current images.
- AI-assisted damage/anomaly detection.
- Change-over-time visualization.
- Heritage Health + Vulnerability score with explainable factors.
- Deterioration/risk trend prediction.
- Scenario simulation for at least one major hazard or environmental stressor.
- Prioritized conservation recommendation.
- Authority dashboard showing what should be inspected first.

10. Ideal Demo Flow

1. Open national/regional map — Show monitored heritage assets and their current risk distribution.
2. Select a heritage site — Open its digital profile and digital twin.
3. Select a component — Example: north façade / wall / foundation.
4. Compare imagery — 2024 vs. 2026; AI highlights newly detected or worsening damage.
5. Explain the risk — Show Health Index, risk drivers and deterioration trend.
6. Predict — Display likely deterioration over the next period under current conditions.
7. Simulate — Show one scenario such as severe rainfall/flood exposure.
8. Recommend — Give a prioritized intervention and suggested inspection window.

11. Suggested Technology Architecture

Layer	Possible technologies
Frontend	Next.js / React, Tailwind CSS
Maps / GIS	Mapbox or Leaflet, OpenStreetMap, PostGIS
3D / Digital Twin	Three.js, GLTF/GLB models
Backend	Python + FastAPI
Computer Vision	OpenCV + pretrained vision models / PyTorch
Prediction	Python ML/time-series models
Data	PostgreSQL/PostGIS + object storage
AI/NLP	Multimodal/LLM layer for report interpretation and structured knowledge extraction

12. Differentiation

Predictive, not reactive: Focuses on forecasting deterioration before irreversible loss.

Living digital twin: The asset is updated as new evidence arrives instead of remaining a static model.

Component-level intelligence: Identifies which part of a structure is deteriorating, not merely that the site is 'at risk'.

Risk-to-action pipeline: Converts detected problems into prioritized inspection and conservation actions.

Multi-factor vulnerability: Treats environmental, structural, human, climate and disaster factors together.

13. Why It Can Work for SIH

Dimension	Why the concept is strong
Novelty	Combines predictive conservation, digital twins, AI visual assessment and heritage intelligence.
Technical depth	Computer vision, GIS, time-series analysis, ML, geospatial data and 3D visualization.
Feasibility	A focused MVP can be demonstrated using one/two sites and controlled datasets.
Practicality	Outputs are directly tied to inspection and conservation decisions.
Impact	Potentially applicable to protected, state-level, local and undocumented heritage.
Scalability	Can expand from a single site to a state/national heritage intelligence network.
User experience	Highly visual: maps, digital twins, damage overlays, risk scores and predictions.

14. Long-Term Vision

The long-term opportunity is a National Heritage Intelligence Network in which every heritage asset has a digital identity; every inspection becomes structured evidence; imagery and environmental observations build a deterioration history; models learn relationships between materials, environment and damage; and conservation interventions are tracked for outcomes.

- IoT sensors for humidity, temperature, vibration, moisture and structural movement.
- Drone-based inspection of inaccessible areas.
- LiDAR/photogrammetry for high-fidelity deformation monitoring.
- AR-assisted field inspection.
- Climate and seasonal risk forecasting.
- A conservation knowledge graph connecting materials, damage mechanisms, causes and successful interventions.
- Crowdsourced citizen observations with human verification.
- National/state-level prioritization of heritage conservation budgets.

15. Risks & What Not to Do

Do not make the project a technology buffet. Avoid presenting AI + IoT + blockchain + drones + AR + VR + satellites as separate innovations. The core story must remain simple and coherent:

Capture condition → create digital twin → detect change → predict deterioration → prioritize intervention.

Disaster resilience remains valuable, but it should be a risk layer within the broader predictive-conservation system. Likewise, citizen reporting, IoT and advanced 3D capture should be presented as extensions rather than mandatory MVP components.

16. Proposed SIH Problem Statement

India's cultural heritage is continuously exposed to structural deterioration, environmental stress, climate variability, human activity and natural hazards. Existing heritage management practices are often fragmented and reactive, relying on periodic inspections and static documentation, making it difficult to identify deterioration trends and prioritize preventive conservation before damage becomes irreversible.

The proposed solution is an AI-powered Heritage Intelligence Platform that creates a dynamic digital twin of heritage assets using historical records, imagery, geospatial information, environmental data and periodic inspections. The platform will detect and track structural and surface-level changes, generate a multidimensional Heritage Health and Vulnerability Index, predict deterioration trends, simulate potential risks, and recommend prioritized conservation interventions.

The system will enable heritage authorities, conservation professionals and researchers to move from reactive damage management to data-driven, predictive and preventive conservation, while providing a scalable framework for monitoring India's diverse and vulnerable cultural heritage.

17. The 20-Second Pitch

"Heritage Shield is a predictive conservation platform that creates a living digital twin of a heritage structure, detects how it is deteriorating, predicts what is likely to happen next, and tells conservation authorities what they should prioritize before the damage becomes irreversible."

18. The Core Product Loop

OBSERVE → images + GIS + historical + environmental data
DIGITISE → digital twin
ASSESS → AI damage detection + Heritage Health

PREDICT → deterioration and vulnerability forecasting
PRIORITISE → asset/component ranking
ACT → recommended conservation intervention
LEARN → monitor outcomes and update the model

Strategic conclusion: Heritage Shield should be presented not as another disaster-management project, but as an early-warning and decision-intelligence system for the survival of heritage itself.